

Underwater WDM-VLC Systems Enabled by Ultrafast Perovskite Superlattice and SiPMs Receiver Array

Zengyi Xu⁽¹⁾, Zhiwu Chen⁽¹⁾, Zhe Feng⁽¹⁾, Zhuoran Hu⁽¹⁾, Yunkai Wang⁽¹⁾, Pengyang⁽²⁾, Congcong Wang⁽²⁾, Jingzhou Li⁽²⁾, Nan Chi⁽¹⁾

⁽¹⁾ Key Laboratory for the Information Science of Electromagnetic Waves (MoE), Fudan University, Shanghai 200433, China; Nanchi@fudan.edu.cn

⁽²⁾ Hangzhou Institute for Advanced Study, University of Chinese Academy of Sciences, No. 1, Sub-Lane Xiangshan, Xihu District, Hangzhou, 310024, China

Abstract We proposed an underwater WDM-VLC system, combining a multi-wavelength laser transmitter and a large area receiver with ultrafast perovskite quantum dot superlattice fluorescent antenna (FA) and SiPM array. The system achieves 117.6 Gbps·mm² capacity-area product in the 1.2 m tap water channel.

Introduction

In the approaching 6G era, the underwater network is an indispensable part of the ubiquitous network[1]. Visible light communication (VLC) has been predicted as a promising technology in highspeed wireless transmission in natural water bodies due to its transmittivity advantage over conventional radio frequency (RF) signals. Besides, underwater VLC (UVLC) systems have other merits such as unpatented spectrum, low electromagnetic leakage, and compatibility to the illumination systems[2]. Therefore, it may play an important role in future underwater links between vehicles, human, platforms, etc.

Recent studies have demonstrated experimental high speed UVLC systems. In these work, WDM-VLC systems show a dominative advantage in channel capacity[3], [4] due to its

multiplication in channel number and transmission rate as well. However, fast as their data rates are, the spot-like receiver has a limited effective area, which increases the difficulty in beam alignment. The highly dynamic characteristic[5] and difficulty in alignment[6] in real underwater environment hinges the collimation of UVLC channels. Increasing the receiver size may alleviate this issue, however, at the cost of lower optoelectrical bandwidth and hence the signal data rate.

In this work, we proposed an underwater WDM-VLC system aimed at multiplying the channel capacity and improving the feasibility of high-speed underwater channel. A 5-wavelength laser (5- λ laser) is adopted as the transmitter and a ultrafast fluorescent antenna (FA) aided 1×4 SiPM receiver array is used as the receiver. According

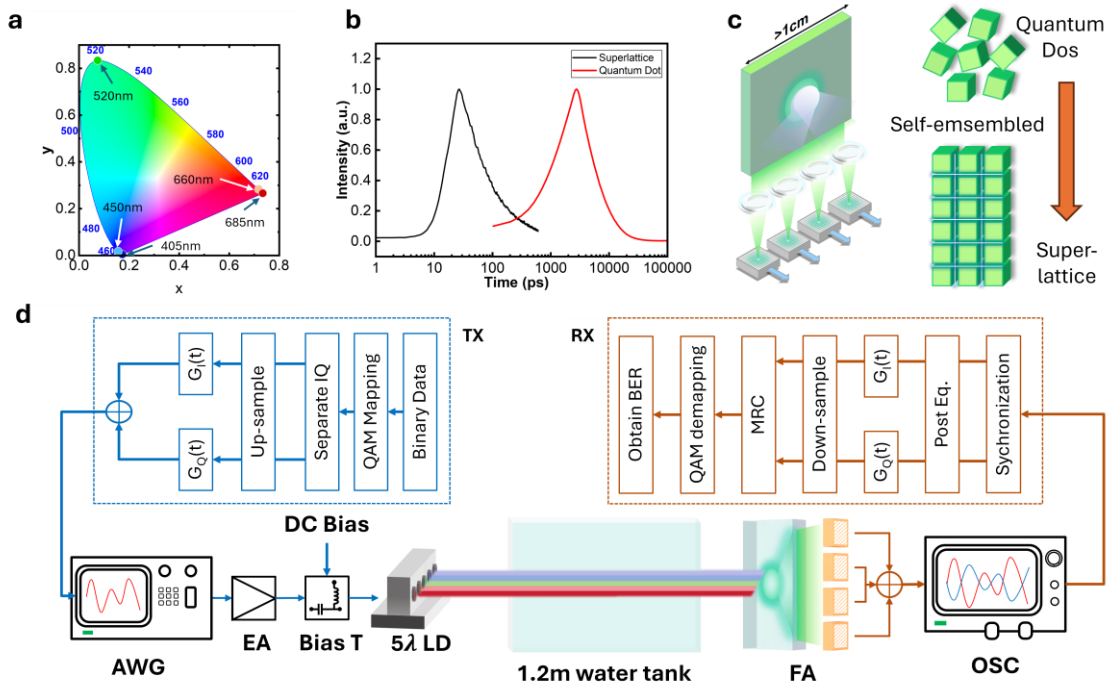


Fig. 1: (a) The chromaticity diagram CIE1931 of the 5- λ laser. (b) The comparison between ultrafast superlattice QD and ordinary QD in fluorescence response. (c) The schematics of the fluorescent antenna (FA) and QD superlattice. (d) The block diagram of the communication system that adopts 5- λ laser and FA.

to the previous study[7], [8], fluorescent antenna is capable to deal with visible light signals with various incident angles. Hence it is useful in mobile communication scenarios between vehicles. Meanwhile, SiPM devices are widely employed in VLC communication[9], [10] and LiDAR applications[11], relying on its high sensitivity and fast response. In our design, the four independent channels of the SiPM array are merged using MRC algorithm. The system achieves 6 Gbps data rate. And the receiver demonstrates a 14 mm by 1.4 mm effective area. The capacity-area product of the system reaches 117.6 Gbps-mm². Compared with conventional point-like PD, it is increased by orders of magnitude.

Device

We proposed a 5- λ laser as the WDM transmitter. Its chromaticity is plotted in Fig.1 (a). The five wavelengths are: 405 nm, 450 nm, 520 nm, 660 nm, and 685nm. The distribution of wavelengths considers the complex environment in underwater channels. The short wavelength channels work with lower losses in clearer waters such as open sea. By contrast, the red channels deal with more turbid water conditions as its longer wavelengths suffers from less scattering.

The receiver adopts superlattice (SL) perovskite quantum dots (QD). Compared with ordinary QD, such material demonstrated ultrafast response speed as shown in Fig. 1(b). It decreases the fluorescent lifetime by two orders of

magnitude, from ns level to ps level. It is caused by the long-range ordered arrangement of the QDs in the SL structure illustrated in Fig. 1(c). Such arrangement causes FRET (Fluorescence Resonance Energy Transfer). It enables excitons to transfer between QDs without causing emission. Therefore, it proves another excessive recombination approach for excitons. Consequently, it generates ultrafast fluorescence to improve the response speed and device bandwidth as well. Nevertheless, even in superlattice QD samples, there are a small fraction of QDs that do not from the superlattice structure and the final response rate is between the measurements of idea superlattice and ordinary QD sample.

Experiment setup

The hardware and software architecture of the underwater communication system are plotted in Fig. 1(b). The system adopts carrierless amplitude and phase (CAP) modulation to modulate complex QAM symbols with pure intensity modulation systems. The CAP waveform is then imported to the arbitrary waveform generator (AWG) Keysight M8190A and the AWG outputs the electric signal to the electric amplifier (EA) Mini-circuit ZHL-1042 J+ (0.01-4.2 GHz). The amplified signal is then coupled with the DC bias using the bias-Tee Mini-circuit AFBT-4R2GW-FT+ (0.1 MHz – 4.2 GHz). The coupled signal drives each individual channel of the 5- λ laser. The emission spectra are shown in Fig. 2(a).

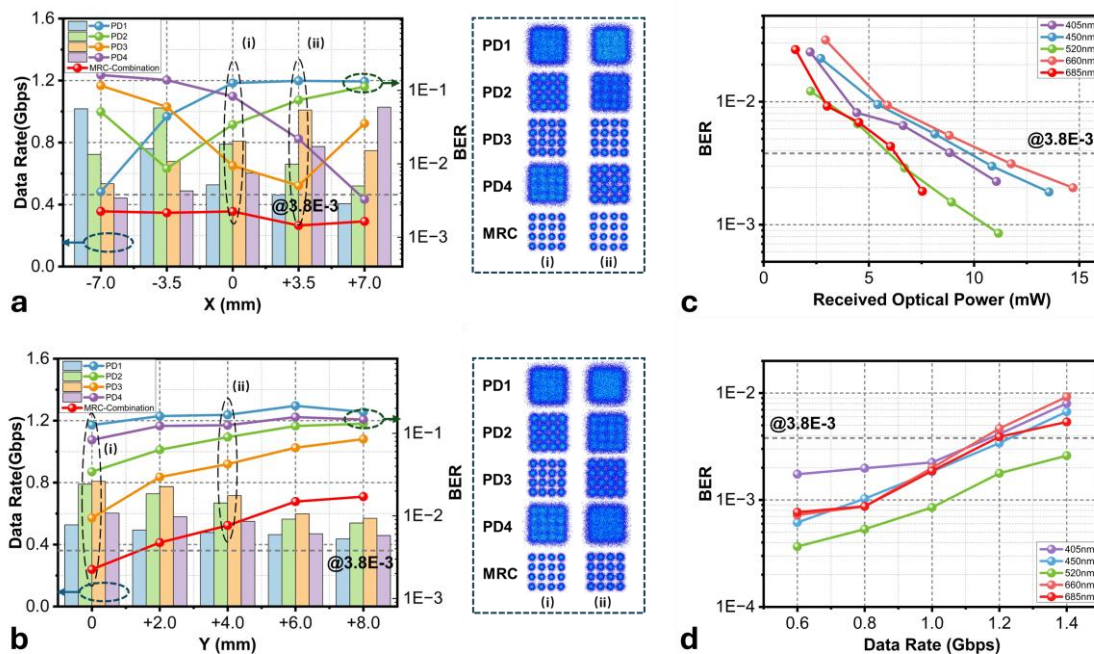


Fig. 2: (a) Estimated data rate (bar chart, left Y-axis) and BER (line chart, right Y-axis) performance of single PD and the MRC mode when shifting the receiver in horizontal direction. (b) BER performance of single PD and the MRC mode when shifting the receiver in vertical direction. (c) The sensitivity experiment result. (d) The maximum channel capacity test result.

The signal penetrates the 1.2 m water tank filled with tap water and reaches the FA-aided SiPM receiver array. The receiver is mounted onto a 2D displacement platform to finely control the horizontal and vertical deviation. The array amplifies the electric signal using the built-in electrical amplifier to enhance the SNR. The four output channels are connected to the channel 1 to 4 of the oscilloscope Keysight MSO9404A, which records the digitized waveform and export data to the computer. Finally, the computer uses MATLAB programs to finish synchronization, post-equalization, down-sampling MRC combination and CAP demodulation.

Result

Firstly, the experiment horizontally moves the receiver and records the estimated accessible data rate and bit error rate (BER) using 405 nm laser. The result is illustrated in Fig. 2(a). The estimated accessible data rate in single PD mode varies oppositely to the distance between the PD and the light beam. For example, the allowed data rate of PD1 is about 400 Mbps when the beam is farthest to it but it quickly increases to approximately 1.0 Gbps when the beam is at top of the PD. Similarly, when fixing the data rate at 1 Gbps, the BER test show that the BER of each PD fluctuates drastically with the horizontal displacement from $1\text{E-}3$ level to $1\text{E-}1$ level. However, with the MRC algorithm, the combined signal exceeds any single PD channels and maintains stably below the BER threshold of 7% HD-FEC ($3.8\text{E-}3$). The stability of signal quality is also supported by the constellation diagrams at the right side of Fig.2(a). The constellation from the MRC mode remains clear and unchanged between area (i) and (ii). When moving the receiver vertically, however, the results differ from the previous one. The charts in Fig. 2(b) do not present a symmetrical pattern because all the receivers are placed at the bottom side. Therefore, when the beam rises from the centre point of the bottom side, the estimated data rate decreases and the BER increases in an almost monotonic way. The maximum data rate drops from 800 Mbps to lower than 600 Mbps for a single PD. And the BER result suggests that the receiver cannot satisfy the FEC requirement once the vertical displacement is more than 4 mm. The constellation diagrams of area (i) and (ii) show a significant deterioration in signal quality and the beam moves away from the receiver array. However, the results in Fig 2 (a-b) proves that the receiver provides a 14mm by 1.4 mm effective area for 1 Gbps VLC channel, which already surpasses most point-like high-speed VLC receivers in effective area.

In the sensitivity test shown in Fig.2(c), we set the data rate at 1 Gbps and vary the optical power by tuning an optical attenuator. The BER

performance changes according to the received optical power (ROP). The sensitivity value is defined as the minimum optical power needed for a successful transmission with BER lower than the threshold. The 520 nm laser channel only require 5.9 mW ROP, presenting the best sensitivity. The sensitivity of 685 nm is 6.3 mW, which is slightly lower than that of 520 nm. 405 nm laser channel 8.8 mW optical signal to satisfy the BER threshold. The lowest sensitivities are those of 450 nm and 638 nm channels. They are 9.8 and 10.7 mW, respectively. The sensitivity measurements do not demonstrate neither positive or negative relationship to the wavelength. It could be caused by the difference in SNR of the laser diodes. All the cases suggest that the system can work with approximately 10 mW ROP for each channel, demonstrating a desirable sensitivity for an overall 5 Gbps underwater communication system.

In the final data rate test, the signal bandwidth is gradually increased from 150 MHz (600 Mbps) to 350 MHz (1.4 Gbps). The BER performances are recorded to indicate the maximum data rate of the corresponding channel. The result is plotted in Fig. 2(d). In this test, the 405 nm and 638 nm channel exceeds the maximum BER allowed when the data rate is 1.2 Gbps. Their highest data rates are hence estimated at 1.1 Gbps. The result of 685 and 450 nm suggest that 1.2 Gbps is at the boundary of their working range. The 520 nm result, however, gives the best data rate as well as it owns the highest sensitivity. It can work at 1.4 Gbps. And finally, the maximum total capacity is estimated at 6 Gbps. Combining the effective area measurement, the system presents a $117.6\text{ Gbps}\cdot\text{mm}^2$ capacity-area product.

Conclusions

We demonstrate an underwater WDM-VLC system with a 5- λ transmitter and an ultrafast super-lattice fluorescent antenna with 1×4 SiPM array structure. The fast response and high sensitivity characteristic enable a desirable data rate and $14\text{ mm} \times 1.4\text{ mm}$ effective area. The total channel capacity is 6 Gbps and the capacity-area product of the system reaches $117.6\text{ Gbps}\cdot\text{mm}^2$, exceeding conventional underwater VLC systems with point-like receivers by orders of magnitude, showing a promising technology for future high-speed underwater VLC systems.

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